PUSHP-RAJ-2024UG3016-(LAB-1)

1. Write a Python program to perform basic arithmetic operations, input/output, and type conversion.

```
a = input("Enter the first number: ")
b = input("Enter the second number: ")

a = int(a);
b = int(b);

print("Sum of the numbers =", a + b)
print("Difference of the numbers =", a - b)
print("Product of the numbers =", a * b)
print("Quotient of a / b =", a / b)
```

```
Enter the first number: 24
Enter the second number: 54
Sum of the numbers = 78
Difference of the numbers = -30
Product of the numbers = 1296
Quotient of a / b = 0.4444444444444444
```

2. Write a Python program using variables, operators, conditional statements, and loops.

```
a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))

c = a + b

if c < 10:
  for i in range(10):
    print(f"{c} * {i+1} =", c * (i+1))
else:
  print(f"{c} * {c} =", c * c)
```

```
Enter the first number: 4
Enter the second number: 65
69 * 69 = 4761
```

3. Write a Python program to demonstrate lists, tuples, sets, and dictionaries.

```
fruits = ['apple', 'mango', 'cherry', 'banana']
print(fruits)
print(fruits[1])

roles = ('student', 'professor', 'worker')
print(roles)
print(roles[1])

roll = {'2024UG3016', '2024UG3016'}
print(roll)

student = {
    "name" : "Pushp",
    "roll" : "2024UG3016"
}
print(student)
```

```
['apple', 'mango', 'cherry', 'banana']
mango
('student', 'professor', 'worker')
professor
{'2024UG3016'}
{'name': 'Pushp', 'roll': '2024UG3016'}
```

4. Write a Python program to create and manipulate NumPy arrays.

```
import numpy as np

a = np.array([1, 2, 3, 4, 5])

print("Original array:", a)

print("First element:", a[0])

a[2] = 10
print("After modification:", a)

print("Array multiplied by 2:", a * 2)

b = np.array([10, 20, 30, 40, 50])

print("Addition:", a + b)
```

```
Original array: [1 2 3 4 5]
First element: 1
After modification: [ 1  2 10  4  5]
Array multiplied by 2: [ 2  4 20  8 10]
Addition: [11 22 40 44 55]
```

5. Write a Python program to create a Pandas DataFrame and perform data selection, filtering, sorting, and aggregation.

```
import pandas as pd

data = {
    "Name": ["Pushp", "Katyayan", "Aayush", "Ujjawal"],
    "Age": [20, 21, 20, 22],
    "Marks": [85, 72, 91, 68],
    "Department": ["DSAI", "DS", "AI", "ECE"]
}

df = pd.DataFrame(data)

print(df)

print("\nSelected columns:")
print(df[["Name", "Marks"]])

print("\nStudents with marks greater than 80:")
print(df[df["Marks"] > 80])

print("\nData sorted by marks:")
print(df.sort_values("Marks", ascending=False))

print("\nAverage marks:", df["Marks"].mean())
print("Maximum marks:", df["Marks"].max())
print("Minimum marks:", df["Marks"].min())
print("Total marks:", df["Marks"].sum())

print("\nAverage marks by department:")
print(df.groupby("Department")["Marks"].mean())
```

```
       Name  Age  Marks Department
0     Pushp   20     85       DSAI
1  Katyayan   21     72         DS
2    Aayush   20     91         AI
3   Ujjawal   22     68        ECE

Selected columns:
       Name  Marks
0     Pushp     85
1  Katyayan     72
2    Aayush     91
3   Ujjawal     68

Students with marks greater than 80:
     Name  Age  Marks Department
0   Pushp   20     85       DSAI
2  Aayush   20     91         AI

Data sorted by marks:
       Name  Age  Marks Department
2    Aayush   20     91         AI
0     Pushp   20     85       DSAI
1  Katyayan   21     72         DS
3   Ujjawal   22     68        ECE

Average marks: 79.0
Maximum marks: 91
Minimum marks: 68
Total marks: 316

Average marks by department:
Department
AI      91.0
DS      72.0
DSAI    85.0
ECE     68.0
Name: Marks, dtype: float64
```

6. Write a Python program to calculate mean, median, and mode for a given dataset.

```python
import pandas as pd

marks = pd.Series([50, 60, 65, 70, 70, 75, 80, 85, 90, 120])

print("Mean:", marks.mean())
print("Median:", marks.median())
print("Mode:", marks.mode()[0])
```

```
Mean: 76.5
Median: 72.5
Mode: 70
```

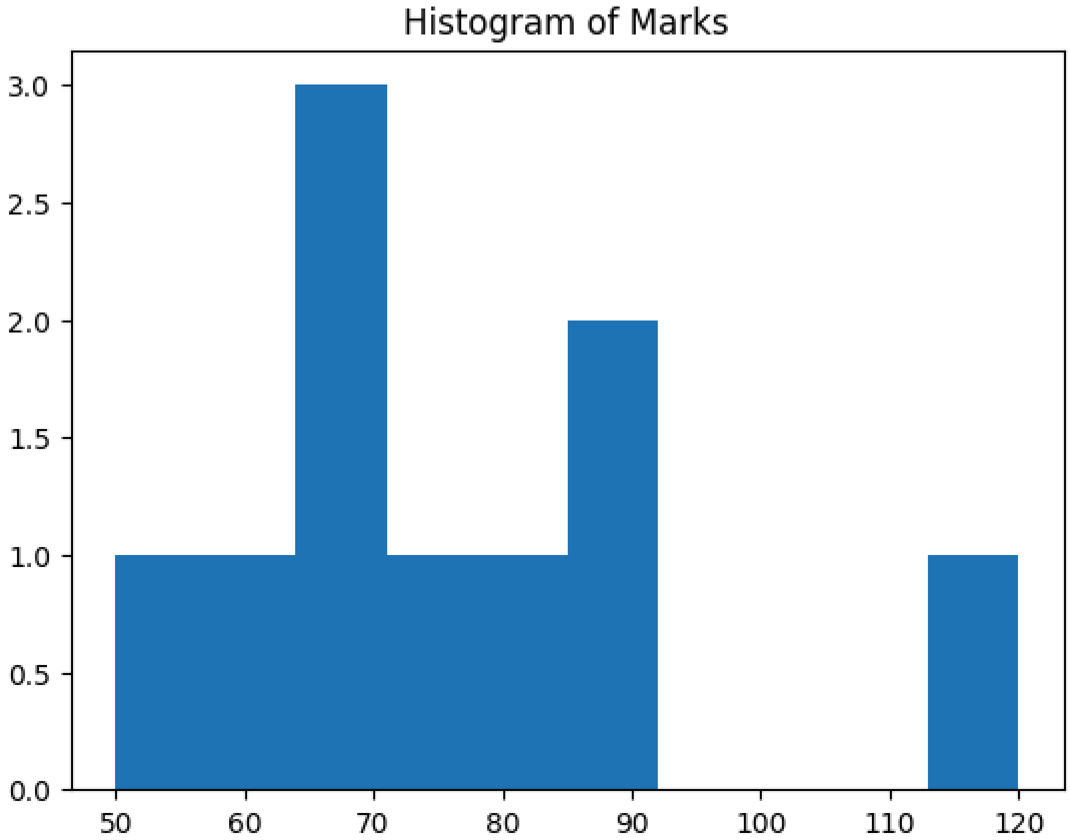
7. Write a Python program to calculate range, variance, standard deviation, quartiles, and interquartile range (IQR).

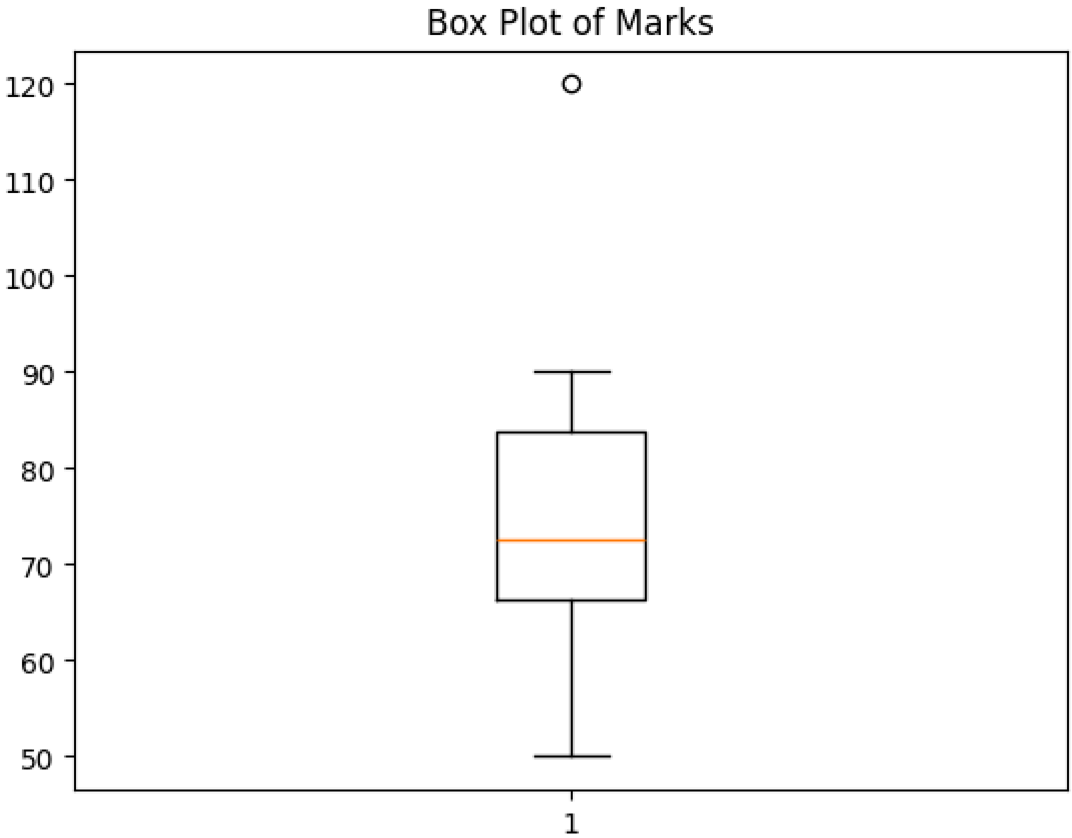
```
import pandas as pd

marks = pd.Series([50, 60, 65, 70, 70, 75, 80, 85, 90, 120])

print("Range:", marks.max() - marks.min())
print("Variance:", marks.var())
print("Standard Deviation:", marks.std())

Q1 = marks.quantile(0.25)
Q2 = marks.quantile(0.50)
Q3 = marks.quantile(0.75)

print("Q1:", Q1)
print("Q2:", Q2)
print("Q3:", Q3)

print("IQR:", Q3 - Q1)
```

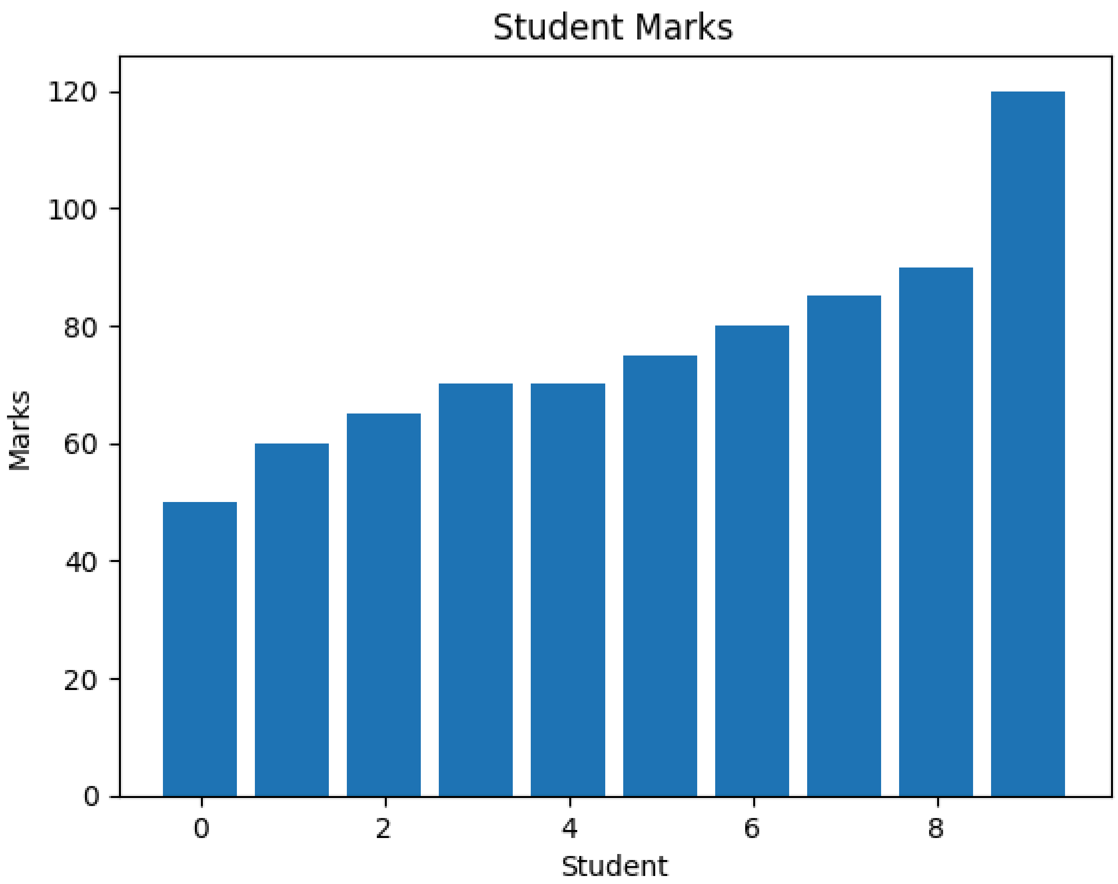
```
Range: 70
Variance: 372.5
Standard Deviation: 19.300259065618782
Q1: 66.25
Q2: 72.5
Q3: 83.75
IQR: 17.5
```

8. Write a Python program to calculate and visualize descriptive statistics using a histogram, box plot, and bar chart.

```
import pandas as pd
import matplotlib.pyplot as plt

marks = pd.Series([50, 60, 65, 70, 70, 75, 80, 85, 90, 120])

plt.hist(marks)
plt.title("Histogram of Marks")
plt.show()

plt.boxplot(marks)
plt.title("Box Plot of Marks")
plt.show()

plt.bar(range(len(marks)), marks)
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.show()
```


Histogram of Marks
3.0
2.5
2.0
1.5
1.0
0.5
0.0
50
60
70
80
90
100
110
120


Box Plot of Marks
120
110
100
90
80
70
60
50
1



9. Given a student marks dataset, calculate mean, median, mode, variance, standard deviation, and identify outliers.

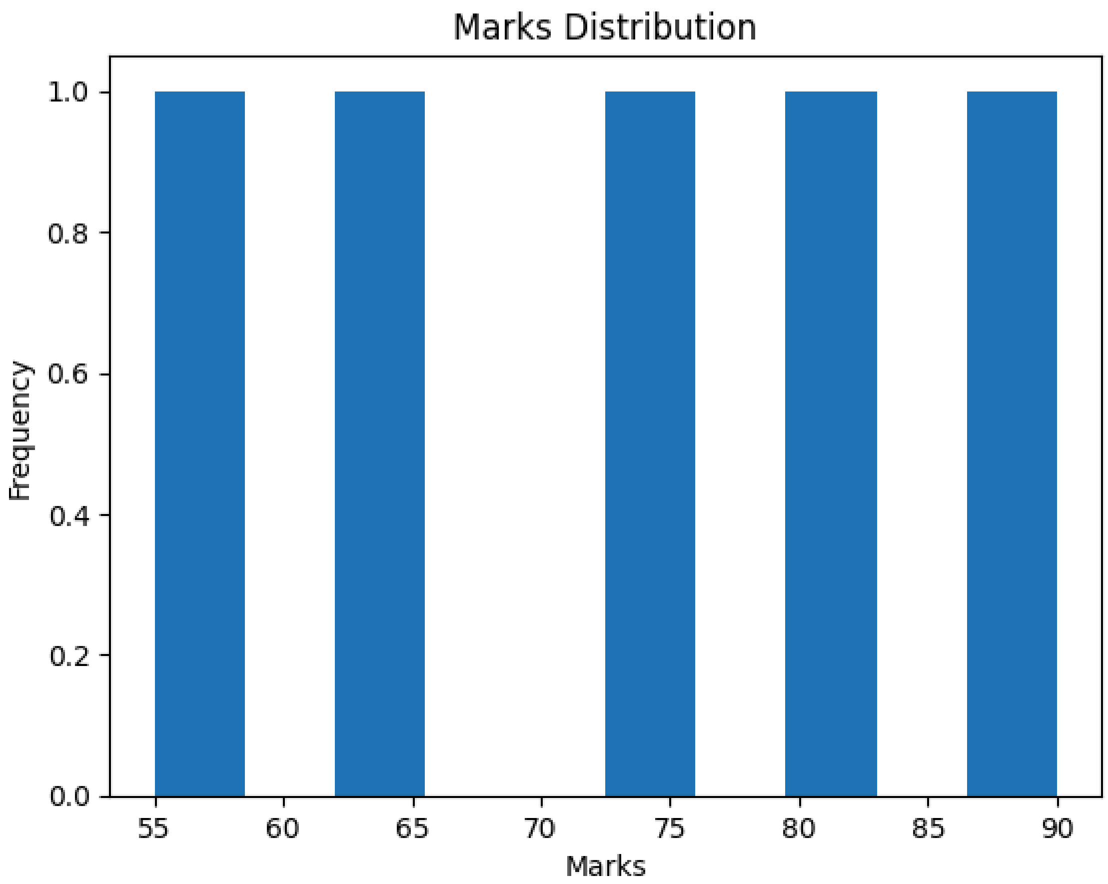
```
import pandas as pd

marks = pd.Series([50, 60, 65, 70, 70, 75, 80, 85, 90, 120])

print("Mean:", marks.mean())
print("Median:", marks.median())
print("Mode:", marks.mode()[0])
print("Variance:", marks.var())
print("Standard Deviation:", marks.std())

Q1 = marks.quantile(0.25)
Q3 = marks.quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("IQR:", IQR)
print("Lower limit:", lower)
print("Upper limit:", upper)

outliers = marks[(marks < lower) | (marks > upper)]

print("Outliers:", list(outliers))
```

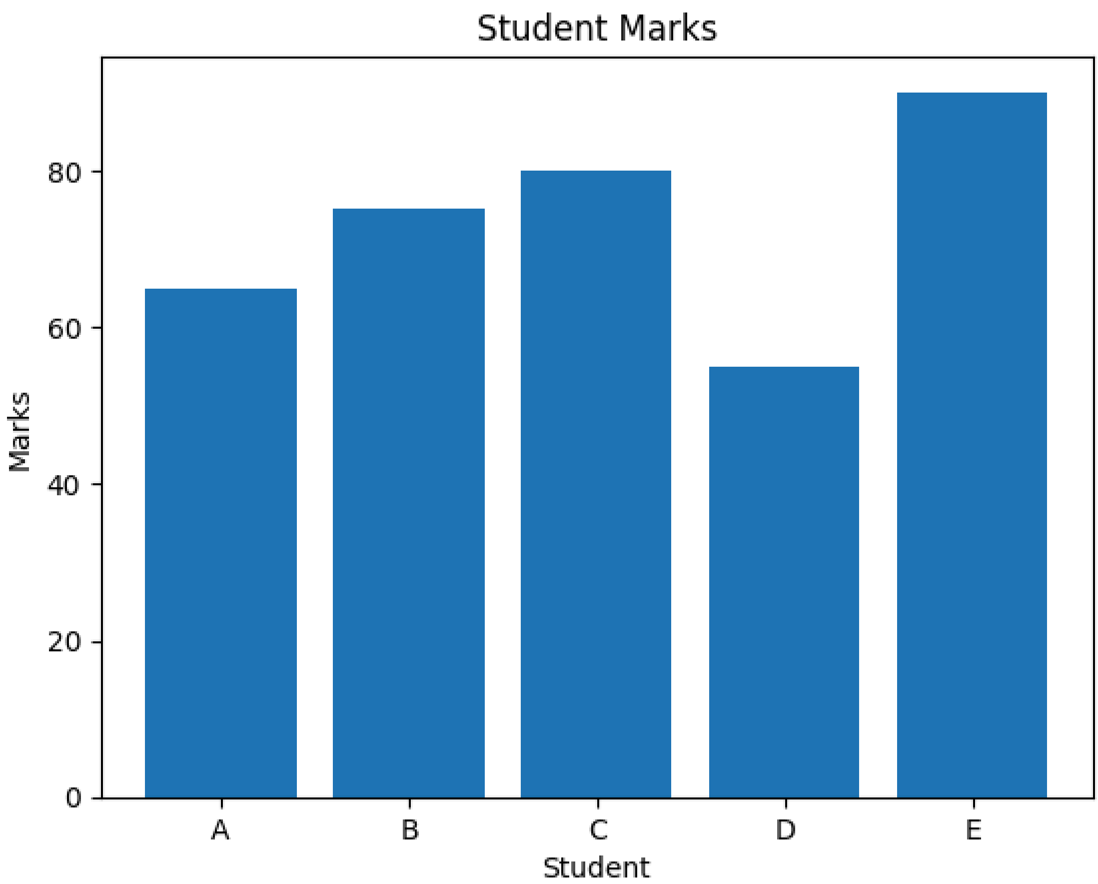
```
Mean: 76.5
Median: 72.5
Mode: 70
Variance: 372.5
Standard Deviation: 19.300259065618782
IQR: 17.5
Lower limit: 40.0
Upper limit: 110.0
Outliers: [120]
```

10. Perform exploratory data analysis (EDA) on a real-world dataset using Pandas, NumPy, and Matplotlib.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = {
    "Name": ["A", "B", "C", "D", "E"],
    "Age": [19, 20, 19, 21, 20],
    "Marks": [65, 75, 80, 55, 90]
}

df = pd.DataFrame(data)

print(df)
print("\nSummary:")
print(df.describe())

print("\nAverage Marks:", np.mean(df["Marks"]))

plt.hist(df["Marks"])
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()

plt.bar(df["Name"], df["Marks"])
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.show()
```

```
  Name  Age  Marks
0    A   19     65
1    B   20     75
2    C   19     80
3    D   21     55
4    E   20     90

Summary:
            Age      Marks
count   5.00000   5.000000
mean   19.80000  73.000000
std     0.83666  13.509256
min    19.00000  55.000000
25%    19.00000  65.000000
50%    20.00000  75.000000
75%    20.00000  80.000000
max    21.00000  90.000000

Average Marks: 73.0
```


Marks Distribution
1.0
0.8
0.6
0.4
0.2
0.0
Frequency
55
60
65
70
75
80
85
90
Marks


Student Marks
80
60
40
20
0
Marks
A
B
C
D
E
Student